# Diffusion models for trajectory planning

Nick Bermbach

## I. Introduction

- Trajectory planning with multiple feasible routes, goals, and obstacle constraints
- Classical trajectory optimization versus learning a distribution of plausible trajectories
- Diffuser's architecture and conditioning, diffusion-based control policies, and a 2D planning study

## II. Problem formulation and diffusion basics

### A. Trajectory planning

- State-action trajectories, system dynamics, and planning horizon; classical trajectory optimization as background [5]
- Start and goal conditions; collision avoidance, path length, and smoothness
- Classical optimization as a comparison: CMA-ES on a parameterized trajectory [2]

### B. Denoising diffusion models

- Forward noising, learned reverse process, and noise-prediction objective (DDPM) [3]
- Distinction between physical trajectory time and diffusion timesteps
- Iterative sampling cost; DDIM as an optional accelerated sampler [7]

## III. Diffuser: learning and conditioning trajectory distributions

### A. Representation, architecture, and training

- Joint state-action trajectories; temporal U-Net and parallel denoising across the planning horizon [4]
- Learning from offline trajectory data; multimodality and dependence on data coverage

### B. Goal conditioning and planning objectives

- Inpainting: fixing the current state and specified goals during iterative denoising
- Reward guidance using a separately learned return predictor
- Planning as inference: $\widetilde{p}_\theta(\tau) \propto p_\theta(\tau) h(\tau)$ [6]
- Maze2D navigation with start-goal inpainting, in contrast to reward-guided locomotion

### C. Constraints and limitations

- Limitations of learned obstacle avoidance and the potential role of explicit cost guidance
- Lack of guarantees for collision avoidance and consistency with system dynamics

## IV. From Diffuser to diffusion policies

### A. Diffusion Policy

- Observation-conditioned generation of action sequences instead of joint state-action trajectories [1]
- Behavior cloning from demonstrations; visual conditioning with temporal CNN or transformer
- Receding-horizon execution and the trade-off between action consistency and responsiveness

### B. One-Step Diffusion Policy

- Distillation of a pretrained diffusion-policy teacher into a single-pass action generator [8]
- Distribution matching using reverse-KL/score-difference objectives
- One-step inference, not one-step training; teacher and distillation costs remain

### C. Comparison of the three methods

- Generated variables, conditioning inputs, training data, and inference procedures
- Differences between trajectory planning and policy learning
- Flexibility, multimodality, runtime, and applicability to the 2D experiment

## V. Experimental study: 2D trajectory planning

### A. Environment and training data

- Point robot with simple velocity-based motion, static obstacles, and start-goal queries
- Diverse, collision-free demonstration trajectories; separate train/validation/test queries
- Comparison with the CMA-ES seminar on a common benchmark

### B. Methods and evaluation

- Simplified Diffuser-style model jointly generating states and actions, with start-goal inpainting
- Inpainting-only baseline versus additional obstacle-cost guidance; CMA-ES on an equivalent task
- Feasibility (collisions, dynamics, speed bound), plan quality, route diversity, and inference runtime

### C. Ablations and failure analysis

- Ablation study: impact of additional obstacle-cost guidance
- Role of training-data coverage; representative collision and dynamics failures
- Limits of conclusions drawn from a simple 2D setting

## VI. Discussion and conclusion

- Comparison of the three papers and the completed 2D experiment
- Trade-offs among data requirements, flexibility, feasibility, solution quality, and computational cost
- Limitations of learned constraints and iterative sampling; open questions for real-world robot control

## References

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